

Curriculum Vitae

Wenli Zhao

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Research Profile

Climate-energy-environment interactions and urban resilience; observation-constrained hybrid Earth-system modeling; ecohydrology and hydroclimate extremes; remote sensing; physics-guided and explainable machine learning; power-system carbon analytics.

Academic Appointments

Spring 2026 Adjunct Assistant Professor, Columbia University, New York, NY
Instructor of record, Hydrosystems Engineering (EACE E3250/E4250), Spring 2026
2025-present Associate Research Scientist, Columbia University, New York, NY
2023-2025 Postdoctoral Research Scientist, Columbia University, New York, NY
Supervisor: Pierre Gentine
2022-2023 Postdoctoral Research Scientist, Max Planck Institute for Biogeochemistry, Jena, Germany
Supervisors: Markus Reichstein and Alexander Winkler
2020-2022 Postdoctoral Research Scientist, Columbia University, New York, NY
Supervisor: Pierre Gentine

Education

2015-2020 Ph.D., Environmental Science, Peking University, Beijing, China
Dissertation: Physics-Constrained Machine Learning of Evapotranspiration; advisors: Guoyu Qiu and Pierre Gentine
2011-2015 B.E., Geographic Information System, Northeast Normal University, Changchun, China
Advisor: Shan Lu

Research Support

Current Award

2025-2028 NASA ROSES-2024 A.63 Ecohydrology, Grant 80NSSC26K0079
Enhancing Mechanistic Understanding of Vegetation-Environment Interactions During Pulse Events with a Differentiable Land Model and Observations | Total award: \$694,276 | Role: Institutional PI/Co-Investigator | Led scientific proposal development and writing; selected as one of 3 awards from 46 submissions

Publications and Manuscripts

Co-first author; * corresponding author; † student-led work advised or co-advised by Wenli Zhao, with contributions to study conception and design. Manuscripts under review or revision are listed separately from published papers.

First- or Corresponding-Author Peer-Reviewed Publications

Zhao, W.*, Winkler, A. J., Reichstein, M., Orth, R., & Gentine, P. (2026). Learning Evaporative Fraction with Memory. Hydrology and Earth System Sciences. In production; Highlight Paper. Preprint: <https://doi.org/10.5194/egusphere-2025-4082> (>4,500 PDF downloads since August 2025).

Zhao, W.*, et al. (2026). Observation-constrained physical snow water equivalent simulations using a physics-guided machine learning approach. Water Resources Research, 62, e2025WR041406. <https://doi.org/10.1029/2025WR041406>

Zhang, S.#†, **Zhao, W.#***, et al. (2025). A near-real time daily European Power Consumption and Carbon Intensity Dataset (ECON-PowerCI). Scientific Data, 12, 1693. <https://doi.org/10.1038/s41597-025-05978-7>

Shi, Z.†, **Zhao, W.***, et al. (2026). A theoretical framework for critical canopy temperature and its application to improve remote-sensing-based estimation of evapotranspiration. Environmental Research Letters, 21, 024011. <https://doi.org/10.1088/1748-9326/ae33d3>

Zhao, W.*, Zhu, B., Davis, S. J., Ciais, P., et al. (2023). Reliance on fossil fuels increases during extreme temperature events in the continental United States. Communications Earth & Environment, 4, 473. <https://doi.org/10.1038/s43247-023-01147-z>. Altmetric attention: 94th percentile among all research outputs tracked by Altmetric.

Zhao, W., Qiu, G. Y., Xiong, Y. J., Gentine, P., & Chen, B. Y. (2020). Uncertainties caused by resistances in evapotranspiration estimation using high-density eddy covariance measurements. Journal of Hydrometeorology, 21, 1349-1365. <https://doi.org/10.1175/JHM-D-19-0191.1>

Zhao, W.*, Gentine, P., Reichstein, M., et al. (2019). Physics-constrained machine learning of evapotranspiration. Geophysical Research Letters, 46, 14496-14507. <https://doi.org/10.1029/2019GL085291>. Recognized as an ESI Highly Cited Paper (top 1% in Geosciences).

Manuscripts Under Review or Revision

- Zhao, W.***, Zhang, S., Zhu, B., et al. (2026). A near-real-time daily European dataset of fuel-specific consumption-based power carbon intensity. Under review.
- Xiong, Y., Jiang, J., Blougouras, G., ... **Zhao, W.*** (2026). Higher vegetation thermal thresholds through adaptation sustain urban cooling. Under review.
- Jiang, J.#, **Zhao, W.#**, Xiao, Z., Yan, C., Mo, D., Jiang, S., & Xiong, Y.* (2026). Estimating daily 10-m land surface temperature by integrating satellite embeddings with physics-guided deep learning. Under review.
- Jiang, J.#, **Zhao, W.#**, Yan, C., Qiu, G. Y., et al. (2026). LoRAET: Natural-to-urban adaptation for daily urban evapotranspiration mapping at 10-m resolution. Under review.
- Zhao, W.***, Zhang, S., Fang, J., et al. (2026). Multi-Scale Memory Governs Ground Heat Flux Dynamics. Under review. Preprint: <https://doi.org/10.2139/ssrn.6852378>
- Ning, X.†, **Zhao, W.***, et al. (2025). Unveiling the carbon-emission mechanism of the global power system under extreme weather: Country-specific insights through explainable machine learning. Under review. Preprint: <https://doi.org/10.2139/ssrn.5405302>
- Cattray, M., **Zhao, W.**, Nathaniel, J., Qiu, J., Zhang, Y., & Gentine, P. (2025). EcoPro-LSTM v0: A memory-based machine-learning approach to predicting ecosystem dynamics across time scales in Mediterranean environments. Under review. Preprint: <https://doi.org/10.5194/egusphere-2024-3726>

Other Peer-Reviewed Publications

- Fang, J.†, Bowman, K., **Zhao, W.**, Lian, X., & Gentine, P. (2026). Differentiable land model reveals global environmental controls on latent ecological functions. *Nature Communications*, 17, 6670. <https://doi.org/10.1038/s41467-026-73395-4>
- Huang, F., Jiang, S., Shangguan, W., Camps-Valls, G., Winkler, A. J., Li, W., Duveiller, G., Reimers, C., **Zhao, W.**, Reichstein, M., & Dai, Y. (2026). Upwind terrestrial influences on soil moisture variability across South America. *Nature Communications*, 17, 7256. <https://doi.org/10.1038/s41467-026-75637-x>
- Jiang, H., Yao, L., Qin, J., **Zhao, W.**, et al. (2026). Building façade photovoltaics enhance global climate resilience. *Nature Climate Change*, 16, 566-574. <https://doi.org/10.1038/s41558-026-02606-z>
- Jiang, H., Yao, L., Qin, J., Bai, Y., Brandt, M., Lian, X., Davis, S. J., Lu, N., **Zhao, W.**, Liu, T., & Zhou, C. (2025). Globally interconnected solar-wind system addresses future electricity demands. *Nature Communications*, 16, 4523. <https://doi.org/10.1038/s41467-025-59879-9>
- Worden, M. A., Bilir, T. E., Bloom, A. A., ... **Zhao, W.**, & Zhu, S. (2025). Combining observations and models: A review of the CARDAMOM framework for data-constrained terrestrial ecosystem modeling. *Global Change Biology*, 31(8), e70462. <https://doi.org/10.1111/gcb.70462>
- Yan, W., Jiang, J., He, L., **Zhao, W.**, et al. (2024). Correcting land surface temperature from thermal imager by considering heterogeneous emissivity. *International Journal of Applied Earth Observation and Geoinformation*, 129, 103824.
- Zhu, B., Deng, Z., Song, X., **Zhao, W.**, et al. (2023). CarbonMonitor-Power: Near-real-time monitoring of global power generation on hourly to daily scales. *Scientific Data*, 10, 217. <https://doi.org/10.1038/s41597-023-02094-2>
- Lian, X., **Zhao, W.**, & Gentine, P. (2022). Recent global decline in rainfall interception loss due to altered rainfall regimes. *Nature Communications*, 13, 7642. <https://doi.org/10.1038/s41467-022-35414-y>
- Yu, L., Qiu, G., Yan, C., **Zhao, W.**, et al. (2022). A global terrestrial evapotranspiration product based on the three-temperature model with fewer input parameters and no calibration requirement. *Earth System Science Data*, 14, 3673-3693. <https://doi.org/10.5194/essd-14-3673-2022>
- Mao, P., Qin, L., Hao, M., **Zhao, W.**, et al. (2021). An improved approach to estimate above-ground volume and biomass of desert shrub communities based on UAV RGB images. *Ecological Indicators*, 125, 107494.
- Hao, M., **Zhao, W.**, Qin, L., et al. (2021). A methodology to determine the optimal quadrat size for desert vegetation surveying based on UAV RGB photography. *International Journal of Remote Sensing*, 42, 84-105.
- Xiong, Y., **Zhao, W.**, Wang, P., et al. (2019). Simple and applicable method for estimating evapotranspiration and its components in arid regions. *Journal of Geophysical Research: Atmospheres*, 124, 9963-9982.
- Yan, C., **Zhao, W.**, Wang, Y., et al. (2017). Effects of forest evapotranspiration on soil water budget and energy flux partitioning in a subalpine valley of China. *Agricultural and Forest Meteorology*, 246, 207-217.
- Lu, S., Lu, X., **Zhao, W.**, et al. (2015). Comparing vegetation indices for remote chlorophyll measurement of white poplar and Chinese elm leaves with different adaxial and abaxial surfaces. *Journal of Experimental Botany*, 66, 5625-5637.

Chinese-Language Peer-Reviewed Publications

- Luo, J., Qin, L., Mao, P., Xiong, Y., **Zhao, W.**, Gao, H., & Qiu, G. (2021). Research progress in retrieval algorithms for chlorophyll-a, a key element of water-quality monitoring by remote sensing. *Remote Sensing Technology and Application*, 36(3), 473-488. In Chinese.
- Zhao, W.**, Xiong, Y., Qiu, G., et al. (2021). Impact of model structure and parameterization differences on evapotranspiration estimation. *Acta Scientiarum Naturalium Universitatis Pekinensis*, 57(1), 162-172. In Chinese.
- Zhao, W.**, Qiu, G., Xiong, Y., et al. (2021). Simulation of sub-daily transpiration characteristics of typical urban trees based on a deep neural network. *Acta Scientiarum Naturalium Universitatis Pekinensis*, 57(2), 322-332. In Chinese.

- Yang, Y., Zou, Z., **Zhao, W.**, & Qiu, G. (2017). Comparative study on the thermal-environment effects of six urban underlying surfaces. *Acta Scientiarum Naturalium Universitatis Pekinensis*, 53(5), 881-889. In Chinese.
- Yang, Y., Zou, Z., **Zhao, W.**, Li, R., & Qiu, G. (2016). Experimental study on surface thermal-environment characteristics of urban pavement bricks using hyperspectral technology. *Ecology and Environmental Sciences*, 25(5), 835-841. In Chinese.
- Yang, T., **Zhao, W.**, Wang, Z., Lu, X., & Lu, S. (2015). Changes in China's cropping system based on remotely sensed NDVI data. *Scientia Agricultura Sinica*, 48(10), 1915-1925. In Chinese.

Selected Presentations

- 2025** How Interpretable Machine Learning Advances Our Understanding of Climate Extremes and Their Impacts on Ecosystems and Infrastructure. AGU Fall Meeting. Oral presentation.
- 2025** Changes in the carbon cycle and its linkage to the water cycle during the 2023-2024 El Niño in the Amazon. AGU Fall Meeting. Poster.
- 2023** An objective estimate of water stress: Going beyond PDSI. EGU General Assembly. Poster.
- 2022** Extreme events increase carbon emissions and reliance on fossil fuels. AGU Fall Meeting. Poster.
- 2022** Physics-constrained machine learning of streamflow. HydroML Symposium. Oral presentation.
- 2020** Physics-constrained machine learning of canopy interception. AGU Fall Meeting. eLightning presentation.
- 2019** Physics-constrained machine learning of evapotranspiration. AGU Fall Meeting. Oral presentation.

Teaching and Mentoring

- Spring 2026** Instructor of Record, Hydrosystems Engineering (EACE E3250/E4250), Columbia University
26 students (14 undergraduate, 12 graduate); responsible for lectures, assignments, in-class activities, and office hours
- 2022** Guest Lecturer, Max Planck Institute for Biogeochemistry
How Extreme Events Challenge Global Power-System Decarbonization
- 2021** Guest Lecturer, Peking University
Machine Learning in Environmental Science: Lessons from Hybrid Modeling of Evapotranspiration
- 2024-present** Research mentor: Julia L. Simpson, Ph.D. candidate, Columbia University
Transfer learning for air-sea flux parameterizations
- 2023-present** Research mentor: Jianing Fang, Ph.D. candidate, Columbia University
Differentiable land modeling and global ecological parameters
- 2023-present** Research mentor: Ning Xuan, graduate student, Peking University
Explainable machine learning of global power-system emissions under extreme weather
- 2022-present** Research mentor: Zhe Shi, Ph.D. candidate, Peking University
Critical canopy temperature and remote-sensing-based evapotranspiration
- 2022-2025** Research mentor: Shujie Zhang, graduate student, Peking University
ECON-PowerCI data development and applications

Professional Service

- Editorial Board member, *Chinese Geographical Science* (2024-present).
- Session chair, AGU 2025: High-Resolution Modeling and Untangling Atmosphere-Hydrology-Ecology Interactions.
- Session chair, AGU 2025: Toward a Hybrid Era of Physical and Machine-Learning Models for Extreme Events.
- Session chair, AGU 2024: Modeling and Untangling Climate-Hydrology-Ecology Interactions Through Extreme Events.
- Judge, Outstanding Student and PhD Candidate Presentation (OSPP) Awards, EGU General Assembly (2023).
- Reviewer for *Nature Geoscience*, *Geophysical Research Letters*, *Water Resources Research*, *Journal of Hydrology*, *Hydrology and Earth System Sciences*, *Agricultural and Forest Meteorology*, *Remote Sensing of Environment*, *JAMES*, *Journal of Geophysical Research: Biogeosciences*, *Geoscientific Model Development*, *Biogeosciences*, and other journals.

Honors and Awards

- 2022** European Commission Seal of Excellence, MSCA Postdoctoral Fellowship proposal
- 2015-2018** First-Class Postgraduate Scholarship, Peking University
- 2016-2017** Learning Excellence Award, Peking University
- 2015-2016** Special Award, Challenge Cup Scientific Research Competition, Peking University
- 2013-2015** Selected member, elite undergraduate class, Northeast Normal University
- 2012-2013** National Students' Innovation and Entrepreneurship Training Program

Technical Expertise

Python, MATLAB, C#, Linux, Google Earth Engine, ArcGIS, ENVI, IDL, GDAL, SQL Server, SPSS, Origin; remote sensing, geospatial analysis, deep learning, hybrid AI-physics modeling, explainable machine learning, process-based land and hydrologic modeling.